

Nikhil Ranjan Tripathi

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Education

Bachelor of Technology in Computer Science

VIT Bhopal University

2023–2027

CGPA: 8.61/10

Senior Secondary Education (Class XII)

Puranchandra Vidyaniketan

2022

Experience

Software Development Intern – Totely India

May 2026–Jul 2026

- Developed CRM dashboards and backend modules for enterprise clients including **3M** and **Carzoo**, streamlining lead management and business workflows.
- Designed REST APIs, CRUD services and optimized database queries to improve application performance and maintainability.
- Built inventory management and reporting modules, providing centralized business analytics and operational insights.
- Collaborated across the software development lifecycle, from requirement analysis and implementation to testing, deployment and production support.

Technical Team Member – iOS Club, VIT Bhopal

2024–2026

- Served on the technical team for two years, contributing to the planning, technical setup and smooth execution of club activities.
- Organized and managed three technical events, including two club-level events and one major institute-level event.

Projects

AI-Powered Smart Housing System

- Developed an AI-powered smart home platform integrating facial authentication and IOT automation.
- Implemented TensorFlow, OpenCV and MQTT for real-time user recognition, appliance control system.
- Built interactive analytics dashboards to visualize energy consumption and overall smart home performance.

AI-Based Smart Fire Evacuation & Dynamic Routing System

- Developed a fire evacuation system using a modified A* algorithm for dynamic route planning in real-time fire risk.
- Simulated temperature, smoke and flame sensor data to calculate node risk scores and update evacuation routes.
- Built a Python Tkinter dashboard with Pushbullet API integration for live monitoring, fire alerts and emergency notifications.

CRM Dashboard System for 3M & Carzoo

- Developed a full-stack CRM platform supporting lead tracking, customer management and workflow automation.
- Implemented authentication, REST APIs and reporting dashboards for efficient business operations.
- Improved operational visibility through centralized analytics, reporting and data-driven decision support.

Technical Skills

Programming Languages: Python, SQL, MATLAB

Web Technologies: HTML5, CSS3, REST APIs

Frameworks & Libraries: TensorFlow, OpenCV, Pandas, NumPy, Scikit-learn

Developer Tools: Git, GitHub, VS Code, Tableau, Power BI, AWS Basics, Jupyter Notebook

Databases: MySQL, SQL

Core Computer Science: Data Structures and Algorithms, Database Management Systems, Operating Systems, Computer Networks, Object-Oriented Programming, Software Engineering

Concepts: Machine Learning, Deep Learning, Statistical Analysis, Data Analysis, Business Intelligence, System Integration, Data Visualization

Certifications

- Machine Learning I – Columbia University
- Google Project Management Specialization – Google
- AWS Technical Essentials – Amazon Web Services
- Google Analytics Certification – Google